

Introduction & Motivation: Problem and Gaps

Problem. Low birth weight (LBW, ≤ 2.5 kg) is linked to elevated neonatal mortality and lifelong morbidity. Incidence reflects interacting socioeconomic, behavioral, and environmental factors. Public-health goal: identify *interpretable* high-risk subgroups for targeted intervention.

Gaps.

1. Classical regression: additive form under-captures interactions.
2. CART: interpretable, but impurity measures (e.g., Gini) ignore full BW distribution; suffers in sparse LBW tails.
3. Bayesian density models: flexible but computationally intensive and opaque.

Contribution. DM-CART: combine CART's interpretability with Bayesian uncertainty by using Dirichlet-Multinomial (DM) *marginal likelihood* as the splitting criterion.

Dataset: Predictors and Coding

The 2021 Natality Data: 3.6 M birth records in all 50 states [5]. Seven predictor variables are chosen and retained based on clinical relevance and known associations with LBW incidence. Predictors are dichotomized into (1/0), yielding $2^7 = 128$ predictor-class combinations.

Table 1. Binary predictor definitions

Label	Field	1 = Yes	0 = No
Boy	sex	Male	Female
Married	dmar	Married	Not married
Black	mrace15	Black / African Am.	Other race
Over33	mager	>33 yr	≤ 33 yr
HighSchool	medu	HS completed	Otherwise
FullPrenatal	prenatal	Adequate care	Inadequate/none
Smoker	cig_0	Any smoking	None

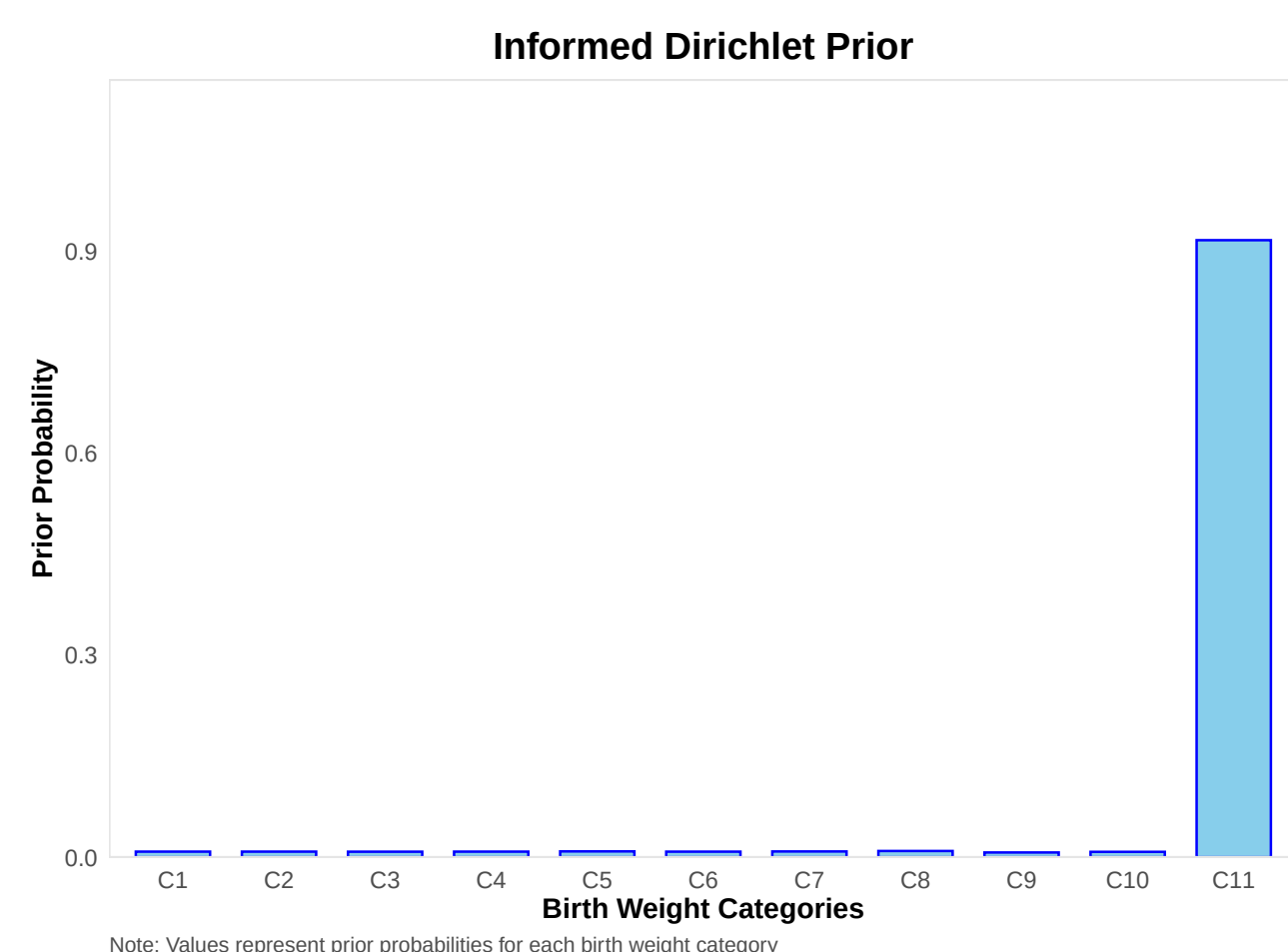
Binning & Priors: Birth-Weight Categories and Dirichlet Parameters

Historic (2020) data define the decline-quantile cut-points in 10% increments in the LBW range (0–2.5 kg), creating **Q1–Q10**, plus NBW (> 2.5 kg).

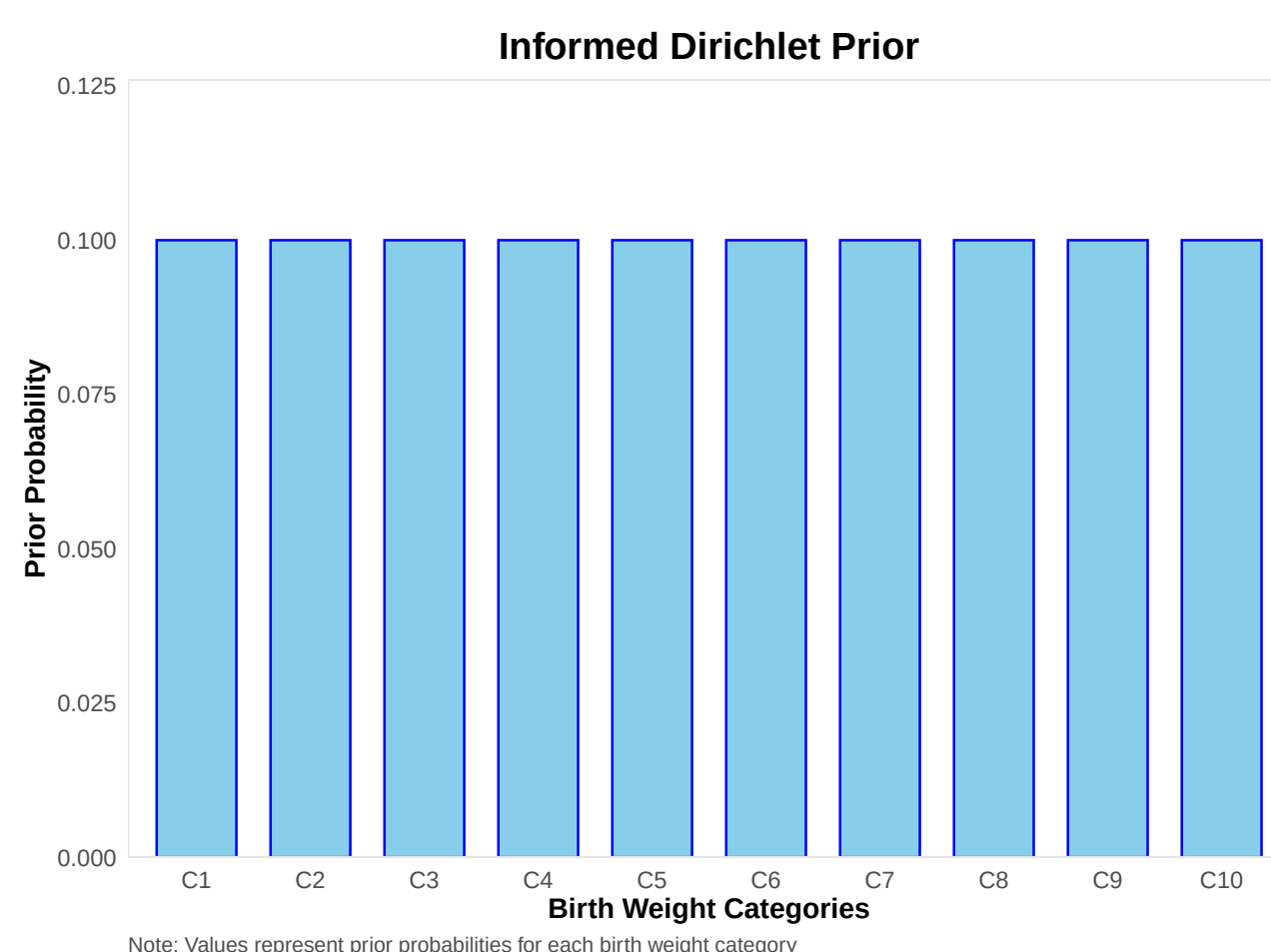
Matrices: 128×11 (full) and 128×10 (LBW-only). $K = \{10, 11\}$ depending on model.

Dirichlet priors $\alpha = (\alpha_1, \dots, \alpha_K)$ from 2020 frequencies:

- Full model: NBW prior $\approx 91.67\%$, LBW split across Q1–Q10.
- LBW-only: uniform 10% per category.



Full model prior: LBW+NBW



LBW-only prior

DM–CART: Splitting Criterion and Likelihood

Predictor: $\mathbf{X} \in \{0, 1\}^{N \times 7}$ matrix and for profile i , row $\mathbf{x}_i \in \{0, 1\}^7$ is fixed.

Response: $\mathbf{Y} = [n_{i,k}]_{i=1}^N$ and for profile i , row $\mathbf{y}_i = (n_{i,1}, \dots, n_{i,K})$ represent BW counts for K total BW categories.

Hierarchical model:

$$\mathbf{y}_i \mid \boldsymbol{\theta}_i, \mathbf{x}_i \sim \text{Multinomial}(N_i, \boldsymbol{\theta}_i), \quad \boldsymbol{\theta}_i \mid \boldsymbol{\alpha} \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_K)$$

where $N_i = \sum_k n_{i,k}$, are the BW counts for profile i , and $\alpha_0 = \sum_k \alpha_k$ is constant.

Note: Multinomial coefficient *omitted* due to data preprocessing and consolidation procedure.

Node log-likelihood:

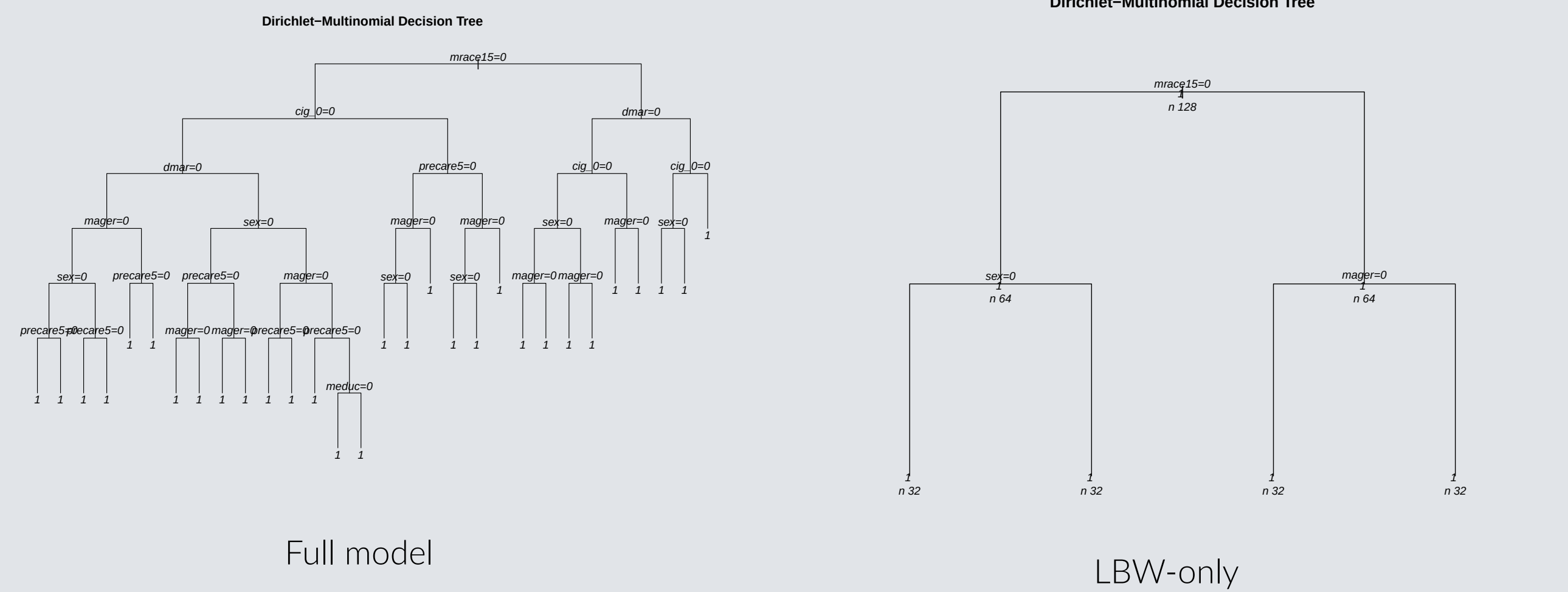
$$\ell_i = \log \Gamma(\alpha_0) - \log \Gamma(N_i + \alpha_0) + \sum_{k=1}^K [\log \Gamma(n_{i,k} + \alpha_k) - \log \Gamma(\alpha_k)]$$

Split gain: $\Delta \ell = \ell_{\text{left}} + \ell_{\text{right}} - \ell_{\text{parent}}$ i.e. improvement in node homogeneity given split [2, 3, 4]

Fitting:

1. Evaluate all candidate splits via $\Delta \ell$
2. Pick max positive split gain (improvement); recurse until depth limit or no-gain.
3. Fit (i) full (NBW+LBW) and (ii) LBW-only models.

Tree Structures: Full vs. LBW-Only Models



Using **rpart** developed by Breiman et al. [1], Therneau et al. [6]

Highlights:

- Full model: Root = race, then smoking, marital status predominantly. *Socioeconomic focus.*
- LBW-only: Root = race, then infant sex, maternal age, prenatal care. *Biological focus.*

Two-Tier Bootstrap: Design and Implementation

Notation: $T = \sum_i \sum_k n_{i,k}$ (grand total), $p_i = N_i/T$, $\mathbf{p} = (p_1, \dots, p_N)$, (empirical class shares)

Tier 1: **Between-class** counts resampling, propagating the prevalence of N profiles.

$\mathbf{n}^* = (n_1^*, \dots, n_N^*) \sim \text{Multinomial}(T, \mathbf{p})$.

Tier 2: **Within-class** counts resampling, for K category counts with $n_i^* > 0$

$\tilde{\mathbf{y}}_i = (\tilde{n}_{i,1}, \dots, \tilde{n}_{i,K}) \sim \text{Multinomial}(n_i^*, \hat{\boldsymbol{\pi}}_i)$, (creating resampled response matrix: $\tilde{\mathbf{Y}}$)

where $\hat{\pi}_{i,k} = (n_{i,k} + \alpha_k)/(N_i + \alpha_0)$ gives the mean bootstrap estimate for (i, k) (Table 1)

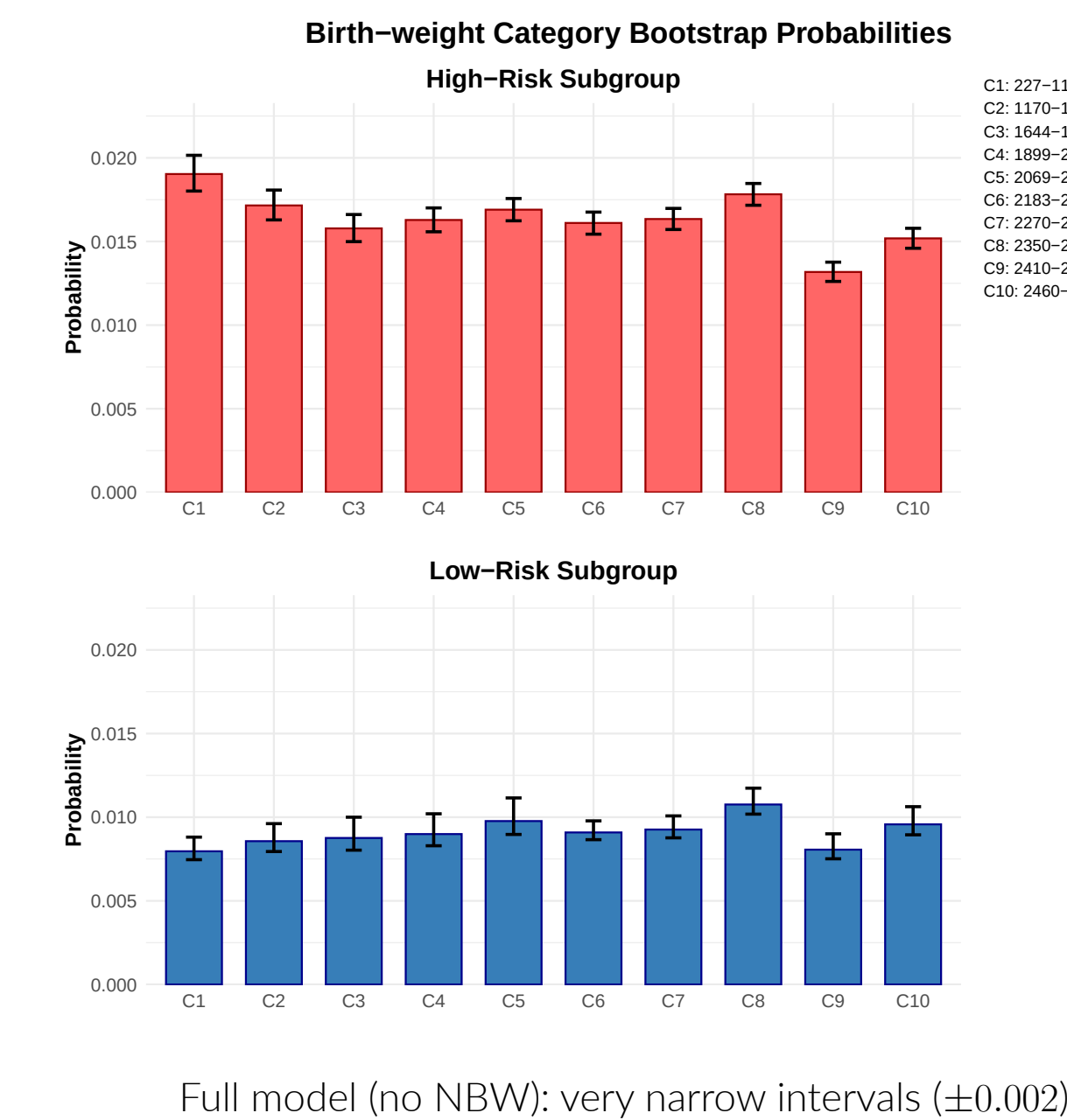
Bootstrap Results: Variable Importance and Interpretation

Variable Importance

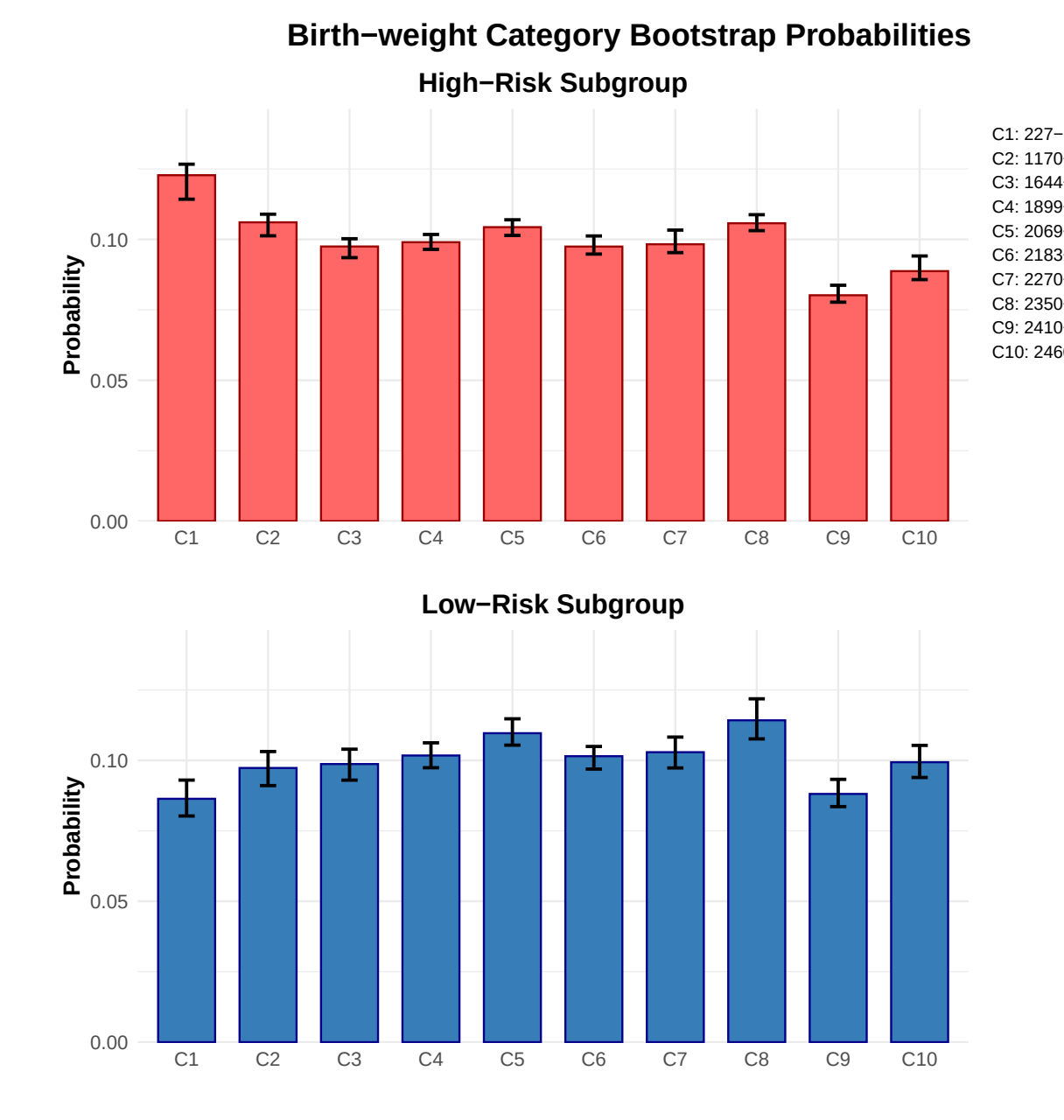
	Full Model	LBW-Only Model
<i>Initial Split Variable</i>		
mrace15	1.0000	1.0000
<i>Variable Frequency</i>		
sex	1.0000	1.0000
dmar	1.0000	0.0915
mrace15	1.0000	1.0000
mager	1.0000	0.9741
precare5	1.0000	0.6351
cig_0	1.0000	0.0196
meduc	0.3794	0.0000

Aggregate mean $\hat{\pi}_{i,k}$ across B bootstraps with 95% percentile intervals.

- High-risk: unmarried, Black, smoking mothers < 33 , $< \text{HS}$ education, inadequate prenatal care, female infant.
- Low-risk: married, non-Black, non-smoking mothers ≥ 33 , $\geq \text{HS}$ education, adequate prenatal care, male infant.
- For NBW, high-risk vs. low-risk prob. gap = 7% (83.6% vs. 90.9%).
- In LBW region, extreme category risk can be 3x higher for high-risk.



Full model (no NBW): very narrow intervals (± 0.002)



LBW-only model: slightly wider intervals (± 0.008)

Conclusions & Future Work

- Model focus:** Full model uses socioeconomic and demographic information to contrast very diverse BW outcomes; LBW-only model identifies diversity within LBW region, biological predictors.
- Key risk factors:** Maternal race, smoking, marital status; refined by infant sex, maternal age, prenatal care.
- Performance:** DM-CART yields clear, epidemiologically consistent subgroup rules, stable across all bootstraps.
- Future work:** Add richer maternal-health & contextual variables, model multi-year trends, and explore clinical EHR integration.

References

- [1] Leo Breiman, Jerome Friedman, R. A. Olshen, and C. J. Stone. *Classification and Regression Trees*. Chapman and Hall/CRC, New York, 1 edition, 1984. doi: 10.1201/9781315139470. URL <https://doi.org/10.1201/9781315139470>.
- [2] Gregory Gundersen. Deriving the dirichlet-multinomial distribution, 2020. URL <https://gregorygundersen.com/blog/2020/12/24/dirichlet-multinomial>. Blog post; accessed 2025-03-03.
- [3] David Mimno. Poly distribution exercise, 2025. URL <https://mimno.infosci.cornell.edu/info6150/exercises/polya.pdf>. Class notes; accessed 2025-03-03.
- [4] Thomas P. Minka. Estimating a dirichlet distribution. Technical report, Microsoft Research, 2000. URL <https://tgminka.github.io/papers/dirichlet/>. Revised 2003, 2009, 2012.
- [5] National Bureau of Economic Research. Vital statistics natality birth data, 2024. URL <https://www.nber.org/research/data/vital-statistics-natality-birth-data>. Data set; accessed 2025-02-23.
- [6] Terry Therneau, Beth Atkinson, and Brian Ripley. rpart: Recursive partitioning and regression trees. 2025. URL <https://cran.r-project.org/web/packages/rpart/rpart.pdf>. Package manual; accessed 2025-03-28.